

NATIONAL UNIVERSITY OF SINGAPORE
CS1101S — PROGRAMMING METHODOLOGY
AY2023/2024 SEMESTER 1



READING ASSESSMENT 2

Time allowed: **40 minutes**

QUESTION PAPER

INSTRUCTIONS

1. This **Question Paper** contains **22 Questions**, and comprises **10 printed pages**, including this page.
2. You are also provided with an **Answer Sheet** to write your answers. It comprises **2 printed pages**.
3. Use a **2B pencil** to **write** your **Student Number** in the designated space on the front page of the **Answer Sheet**, and **shade** the corresponding circle **completely** in the grid for each digit or letter. **DO NOT WRITE YOUR NAME.**
4. You must **submit only** the **Answer Sheet** and no other documents.
5. All questions must be answered in the space provided in the **Answer Sheet**; no extra sheets will be accepted as answers.
6. For **multiple choice questions (MCQ)**, use a **2B pencil** to **shade** in the **circle** of the correct answer **completely**.
7. Each question has one correct answer. The **indicated marks** are awarded for each correct answer and there is **no penalty** for a wrong answer.
8. The **full score** of this assessment is **60 marks**.
9. Answer **all questions**.
10. This is a **closed-book** assessment.

- (1) [1 mark] What is the **alphabet letter** that is in the box at the **top-right corner** on the **front page** of this **question paper**? (**Important:** Please make sure your answer is correct because it determines how your answers to all the subsequent questions will be graded.)

- A. A
- B. B
- C. C
- D. D

Section A [19 marks]

- (2) [3 marks] What is the result of evaluating the following Source §3 program?

```
function f(g) {
    return g(x);
}
{
    const x = 1;
    f(y => y + 2);
}
```

- A. undefined
- B. Error: Name x not declared.
- C. Error: Name y not declared.
- D. 1
- E. 3
- F. 2
- G. None of the other options is the correct answer

- (3) [3 marks] What is the result of evaluating the following Source §3 program?

```
let f = x => x + 1;
function g(x) {
    return y => f(x);
}
const h = g(2);
f = x => x + 3;
h(4);
```

- A. None of the other options is the correct answer
- B. 2
- C. 1
- D. 4
- E. 5
- F. 6
- G. 3

(4) [3 marks] What is the result of evaluating the following Source §3 program?

```
let x = 1;
function f(y) {
  x = 2;
  let x = 8;
  return x + y;
}
f(4) + x;
```

- A. 12
- B. 13
- C. 14
- D. Error: Cannot access 'x' before initialization.
- E. None of the other options is the correct answer
- F. 7
- G. 8
- H. 6

(5) [3 marks] How many **function objects** get created during the evaluation of the following Source §3 program? (Do not count function objects of primitive and pre-declared functions.)

```
function f(x) {
  return x === 1
    ? y => 10
    : x === 2
    ? y => 20
    : x === 3
    ? y => 30
    : x === 4
    ? y => 40
    : y => 50;
}
f(1)(5);
```

- A. 3
- B. More than 5
- C. 2
- D. 5
- E. 0
- F. 4
- G. 1

(6) [4 marks] What is the result of evaluating the following Source §3 program?

```
let f = x => {
  f = y => y + 2;
  return f(x) + 3;
};
f(1);
f(1);
```

- A. Error: The function has encountered an infinite loop.
- B. Error: Cannot assign new value to constant f.
- C. 3
- D. 6
- E. 7
- F. 4
- G. None of the other options is the correct answer
- H. 5

(7) [3 marks] What is the result of evaluating the following Source §3 program?

```
let f = b => (f = b => !b)(!b);
while (f(true)) {
  f(true);
}
```

- A. true
- B. Error: The loop has encountered an infinite loop.
- C. Error: The function has encountered an infinite loop.
- D. b => !b
- E. undefined
- F. None of the other options is the correct answer
- G. false
- H. Error: Cannot assign new value to constant f.

Section B [22 marks]

For all the questions in this section, consider the following Source §3 program:

Program B:

```
// xs and ys must be lists of the same length
function fun(xs, ys) {
  if (is_null(xs)) {
    return null;
  } else {
    set_tail(ys, fun(tail(xs), tail(ys)));
    const new_pair = pair(head(xs), ys);
    return new_pair;
  }
}
const PP = list(list(1), list(2), list(3));
const QQ = list(4, 5, 6);
const RR = fun(PP, QQ);
```

(8) [2 marks] What is the value of RR (in *list notation*) at the end of the evaluation of Program B?

- A. list(1, 2, 3)
- B. list(list(1), 4, list(2), 5, list(3), 6)
- C. list(list(1), list(2), list(3))
- D. list(list(1), list(4), list(2), list(5), list(3), list(6))
- E. list(1, 4, 2, 5, 3, 6)
- F. None of the other options is the correct answer
- G. list(list(1), 4, 5, 6)
- H. list(1, list(4), 2, list(5), 3, list(6))

(9) [2 marks] What is the value of PP (in *list notation*) at the end of the evaluation of Program B?

- A. list(1, 4, 2, 5, 3, 6)
- B. list(1, 2, 3)
- C. list(list(1), list(4), list(2), list(5), list(3), list(6))
- D. list(list(1), list(2), list(3))
- E. list(list(1), 4, 5, 6)
- F. list(1, list(4), 2, list(5), 3, list(6))
- G. list(list(1), 4, list(2), 5, list(3), 6)
- H. None of the other options is the correct answer

(10) [3 marks] What is the value of QQ (in *list notation*) at the end of the evaluation of Program B?

- A. `list(4, list(2), 5, list(3), 6)`
- B. `list(list(4), 2, list(5), 3, list(6))`
- C. `list(list(1), 4, list(2), 5, list(3), 6)`
- D. None of the other options is the correct answer
- E. `list(4, 5, 6)`
- F. `list(4, 2, 5, 3, 6)`
- G. `list(list(4), list(5), list(6))`
- H. `list(list(4), list(2), list(5), list(3), list(6))`

(11) [3 marks] What is the result of the following statement if it is evaluated at the end of Program B?

```
[ PP === RR, head(PP) === head(RR), tail(PP) === tail(RR) ];
```

- A. `[false, false, false]`
- B. `[false, false, true ]`
- C. `[false, true, false]`
- D. `[false, true, true ]`
- E. `[true, false, false]`
- F. `[true, false, true ]`
- G. `[true, true, false]`
- H. `[true, true, true ]`

(12) [3 marks] How many **pairs** get created during the evaluation of Program B?

- A. 12
- B. 24
- C. None of the other options is the correct answer
- D. 3
- E. 9
- F. 6
- G. 18
- H. 15

(13) [2 marks] How many **bindings** does the **program environment frame** contain during the evaluation of Program B?

- A. 7
- B. 2
- C. 5
- D. 1
- E. More than 7
- F. 6
- G. 3
- H. 4

(14) [4 marks] How many **environment frames** get created during the evaluation of Program B? (Do not count the global environment frame. We assume that the applications of the primitive functions, such as `is_null`, `head`, `tail`, `pair`, `list`, `set_head`, and `set_tail`, do not create any frame.)

- A. 7
- B. Fewer than 4
- C. 9
- D. 4
- E. More than 9
- F. 8
- G. 6
- H. 5

(15) [2 marks] Of the **environment frames** that get created during the evaluation of Program B, how many **extend the program environment *directly***? (We assume that the applications of the primitive functions do not create any frame.)

- A. 0
- B. 4
- C. More than 6
- D. 2
- E. 5
- F. 1
- G. 6
- H. 3

(16) [1 mark] How many **function objects** get created during the evaluation of Program B? (Do not count function objects of primitive and pre-declared functions such as `array_length`, `math_floor`, `pair`, `head`, `tail`, `length`, and `map`.)

- A. More than 6
- B. 4
- C. 3
- D. 1
- E. 2
- F. 5
- G. 0
- H. 6

Section C [18 marks]

For all the questions in this section, consider the following Source §3 program:

Program C:

```
let A = [2, 1];
function check(A) {
  let i = 0;
  let sum = 0;
  while (i < array_length(A)) {
    let curr = A[i];
    sum = sum + curr;
    i = i + 1;
  }
  return () => sum > A[i - 1];
}
let curr = 0;
while (check(A)()) {
  A[array_length(A)] = curr;
  curr = curr - 1;
}
[curr, array_length(A)];
```


(17) [4 marks] What is the result of evaluating Program C?

- A. [-4, 7]
- B. [-3, 6]
- C. [-5, 8]
- D. None of the other options is the correct answer
- E. [-4, 6]
- F. [-3, 5]
- G. [-5, 7]
- H. [-3, 7]

(18) [4 marks] How many **environment frames** get created during the evaluation of Program C? (Do not count the global environment frame. We assume that the applications of the primitive functions, such as `array_length`, do not create any frame.)

- A. 16
- B. 40
- C. 9
- D. 12
- E. 14
- F. More than 40
- G. 31
- H. 23

(19) [3 marks] Of the **environment frames** that get created during the evaluation of Program C, how many **extend the program environment *directly***? (We assume that the applications of the primitive functions do not create any frame.)

- A. 9
- B. 3
- C. 8
- D. 4
- E. More than 9
- F. 5
- G. 6
- H. 7

(20) [2 marks] Of the **environment frames** that get created during the evaluation of Program C, how many extend directly from the environment frame in which the binding for the variable **i** has the final value 4?

- A. 5
- B. 3
- C. 4
- D. 8
- E. 2
- F. 7
- G. More than 8
- H. 6

(21) [2 marks] How many **bindings** does the **program environment frame** contain during the evaluation of Program C?

- A. More than 7
- B. 1
- C. 7
- D. 2
- E. 3
- F. 5
- G. 4
- H. 6

(22) [3 marks] How many **function objects** get created during the evaluation of Program C? (Do not count function objects of primitive and pre-declared functions.)

- A. 6
- B. 2
- C. 9
- D. 1
- E. 5
- F. 7
- G. 3
- H. More than 9

———— **END OF QUESTIONS** ————